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Separation Distress and Attachment in Surrogate-Reared Squirrel Monkeys[1]

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HENNESSY, M. B., J. N. KAPLAN, S. P. MENDOZA, E. L. LOWE AND S. LEVINE. *Separation distress and attachment in surrogate-reared squirrel monkeys.* PHYSIOL. BEHAV. **23**(6) 1017–1023, 1979.—Surrogate-reared infant squirrel monkeys were exposed to various conditions of separation from their surrogate. Infants showed significant increases in plasma levels of cortisol when they were placed in an unfamiliar environment during the separation period. Changes in behavior, but not cortisol, were observed under conditions in which the surrogate was removed and the infant left in the home cage. These results differ from those previously obtained with mother-reared infants. It is concluded that surrogate-reared infant squirrel monkeys do not show the same separation response or attachment to their rearing figure as do mother-reared infants.

Attachment Mother-infant separation Cortisol response Surrogate rearing Squirrel monkey Surrogate "aunts"

THE existence of strong emotional bonds, or attachement, between mother and infant has been described in a number of primate species, including humans [3,27]. One defining characteristic of attachment is the occurrence of distressful reactions upon involuntary separation from the attachment figure [1]. In both human and nonhuman primates, removal of the mother results in an immediate increase in agitation or 'protest' behaviors in the infant [11, 12, 13, 20, 33]. If separation continues, the initial protest stage is sometimes followed by a second stage of behavioral suppression or 'despair' [20,31].

Although investigations of mother-infant separation have traditionally focused on behavioral responses, recent studies have shown that separation can result in physiological indicants of increased stress. In one study [2], it was found that a 6-day separation of infant rhesus monkeys from their mothers produced enzymatic activity indicative of increased sympathetic action and an increase in brain serotonin levels. Other investigators have noted changes in body temperature, heart rate and sleep patterns which correlate with the behavioral changes seen during the protest and despair stages of separation in pigtail monkey infants [28,30]. Further, Smotherman *et al.* [34] found that infant rhesus monkeys which had been separated from their mothers for 3 hr showed significant elevations of plasma levels of cortisol. Separation-induced increases in pituitary-adrenal activity have also been observed in a New World primate, the squirrel monkey. We have found that a 30-min period of separation produces a marked increase in the plasma cortisol levels of both infant and mother [25]. However, if the infant is removed from the mother but then immediately reunited with her, no such increases are observed [22,25]. In our initial work, animals were always put into an unfamiliar room during separation so we could determine the extent to which the cortisol response was due to exposure to strange surroundings, rather than to separation per se. However, Coe *et al.* [4] showed that when either infants or their mothers were left in the home cage with other familiar animals during the separation period, the cortisol response was as great as when the infants or mothers were placed in an unfamiliar environment during separation. Thus, the cortisol response to the separation procedure in these infants and their mothers could be entirely accounted for by the absence of the attachment object.

Infant rhesus monkeys have also been found to cling and show apparently normal attachment to cloth-covered artificial mothers (surrogates) [6,7]. When separated from the surrogate and placed into an unfamiliar environment, the infant shows emotional behavioral reactions [6,7]. Further, Hill *et al.* [10] found that when rhesus infants were separated from their rearing surrogate and placed in an unfamiliar environment, they showed a greater cortisol response than when

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they were placed in the unfamiliar environment with their surrogate. However, more recent evidence suggests that attachment of macaque infants to a surrogate differs from attachment to a biological mother. Meyer *et al.* [26] found that during a 9-week separation from their rearing figures, surrogate plus peer-reared rhesus infants showed less behavioral signs of distress as well as lower resting levels of plasma cortisol and a smaller cortisol response to the manipulation of a nearby infant than did mother-reared infants. Thus it appeared that separation was less traumatic for the surrogate plus peer-reared group. Similarly, Reite *et al.* [29] found that changes in body temperature, heart rate, sleep patterns, as well as behavior following separation were less intense in surrogate-reared pigtail monkey infants than had previously been observed in their mother-reared counterparts [28,30].

In our previous work with the squirrel monkey, we found that separation into an unfamiliar room produced as great an elevation of plasma cortisol levels in surrogate-reared as it did in mother-reared infants [25]. However, whereas Coe *et al.* [4] showed that the cortisol response of mother-reared infants could be attributed to separation per se, the differential effects of separation from the rearing surrogate and placement into an unfamiliar environment on the cortisol response of surrogate-reared infants could not be assessed from our initial studies. If attachment to an inanimate surrogate and to a biological mother is equivalent for the infant squirrel monkey, one would expect infants to show an increase in plasma cortisol concentrations following separation from the surrogate, regardless of whether or not the procedure involved removing the infant to a strange environment. The present study investigated this issue.

EXPERIMENT 1

In this experiment, we exposed surrogate-reared infant squirrel monkeys to various separation procedures in order to determine if the cortisol response to surrogate separation [25] results from separation per se, or from exposure to an unfamiliar environment during the separation period. The degree to which experimenter handling of the infants contributes to the cortisol response was also investigated.

METHOD

Animals

Four male and five female infant squirrel monkeys of the Bolivian variety were used in this experiment. They were between 18 and 20 weeks old at the beginning of the experiment and had been reared on inanimate surrogates from approximately 1 wk of age. The physical characteristics of the surrogates as well as the rearing procedures have been described in detail in previous reports [14,17]. In brief, infants were housed individually in cages (61×46×41 cm) with their surrogate, which consisted of a plastic cylinder (5 cm dia. ×25 cm long) covered with an acrylic "fur" material. All animals were maintained in the same large room, which also contained other squirrel monkeys. A 12-hr light/dark cycle was imposed (lights on 0700 hr).

Experiment Conditions

Each infant was exposed to the following five experimental conditions, with individual conditions presented at weekly intervals. Order of condition presentation was counterblanced across infants.

(1) Basal. In this condition, the infant was rapidly removed from its cage, anesthetized with ether and a 0.5 ml sample of blood collected via cardiac puncture [25]. This condition was used to estimate the infants' resting levels of plasma cortisol.

(2) Separated-Unfamiliar Environment-Handled. The infant was removed from its surrogate and placed alone in a cage similar to its home cage in a different room for 30 min. The unfamiliar room, like the rearing room, contained other monkeys of various age/sex classes that could be seen and heard but not contacted by the separated infant. At the end of the 30-min separation period a blood sample was collected. This condition assessed the combined effects of handling the infant, separating it from its surrogate and placing it in a unfamiliar environment, and was essentially the same as has previously been shown to result in elevated plasma cortisol levels [25].

(3) Separated-Familiar Environment-Nonhandled. The surrogate was removed from the home cage with as little disturbance to the infant as possible, while the infant remained in the cage. Thirty minutes following separation, a blood sample was obtained. Thus, this condition determined the effect on plasma cortisol levels of separation from the surrogate without exposure to an unfamiliar environment or handling of the infant.

(4) Nonseparated-Familiar Environment-Handled. The infant was removed from its surrogate and manually held outside of its cage for the same length of time as it took to transfer the infant from its home cage to the unfamiliar cage in Condition 2. After being handled, the infant was returned to its surrogate in the home cage and a blood sample was collected 30 min later. This condition determined the effect of experimenter handling alone on plasma cortisol levels of the infant.

(5) Separated-Familiar Environment-Handled. The infant was removed from its cage and manually held for the same length of time as in Conditions 2 and 4. The infant's surrogate was removed while it was being held and then the infant was returned to its empty cage for 30 min, after which a blood sample was obtained. This condition was included to assess the combined effects of surrogate separation and handling on the cortisol response of the infant.

Cortisol Determinations

Blood samples were centrifuged following collection and the obtained plasma was frozen until assay. Cortisol was assayed using antiserum No. F2153 from Endocrine Sciences, Tarzana, California, as described by Klemm and Gupta [21]. Details of the assay have been reported previously [25]. Triplicate aliquots were assayed from each sample and averaged.

RESULTS

A one-way analysis of variance for repeated measures revealed a significant overall effect, $F(4,32)=9.20$, $p<0.001$ (Fig. 1). A Newman-Keuls post hoc test showed that the cortisol levels obtained for the *Separation-Unfamiliar Environment-Handled* condition were significantly higher than values obtained during any of the other four conditions ($ps<0.01$). No other differences between pairs of means were significant.

EXPERIMENT 2

In Experiment 1 we found that surrogate-reared infant

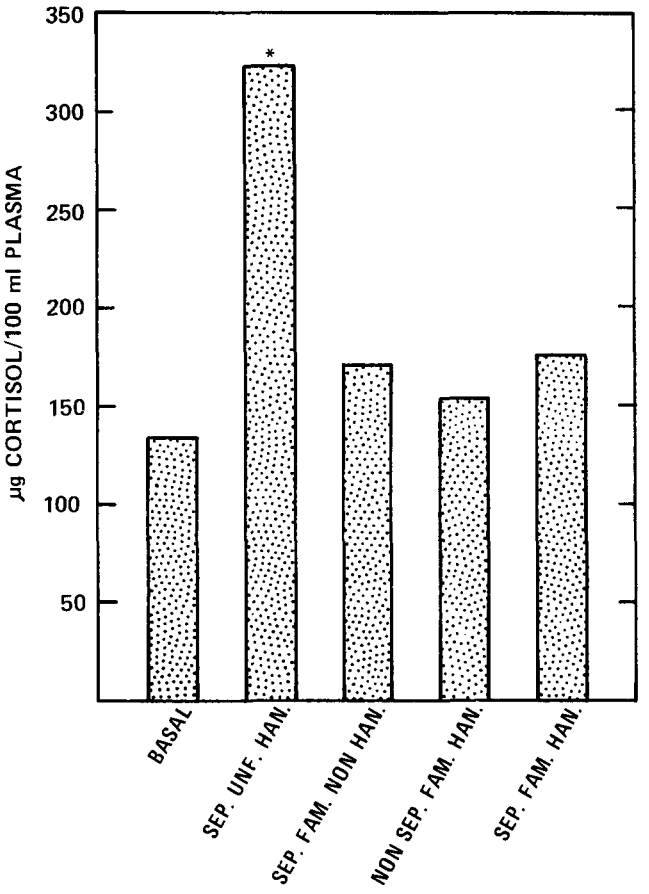


FIG. 1. Plasma cortisol levels of infant squirrel monkeys in each of the five experimental conditions of Experiment 1: *Basal; Separated-Unfamiliar Environment-Handled* (SEP. UNF. HAN.); *Separated-Familiar Environment-Nonhandled* (SEP. FAM. NON HAN.); *Nonseparated-Familiar Environment-Handled* (NON SEP. FAM. HAN.); *Separated-Familiar Environment-Handled* (SEP. FAM. HAN.). *Differs from all other means, $p<0.01$.

squirrel monkeys responded to separation from the surrogate with an increase in plasma levels of cortisol only if the separation procedure involved removing the infant to an unfamiliar environment during the separation period. If the surrogate was simply removed from the infant's home cage, or if the infant was handled either with or without surrogate removal, no increase in plasma cortisol was observed. These results differ from those obtained with mother-reared infants. Coe *et al.* [4] found that infants showed a cortisol response to separation from their biological mother, even if they were left in the familiar home cage during the separation period. Thus, it seems that surrogate rearing alters the response of infant squirrel monkeys to separation. However, other differences between Experiment 1 and Coe *et al.* [4] may also explain the data. In the Coe *et al.* study, infants were left with other animals of their social group during the separation period, and three of the four infants were immediately 'aunted', i.e., picked up and carried in the typical infant position on the back of a nonlactating female. Although aunts can reduce behavioral signs of infant distress [32], it has been shown that infant squirrel monkeys can distinguish and show a clear preference for their mother over other females [15]. Therefore, it is possible that the cortisol elevation observed by Coe *et al.* resulted to some degree from the infant's detection that the female to which it was clinging was not its mother. In Experiment 1, the absence of potential aunts during the separation period precluded the possibility of this occurrence.

A second difference between the Coe *et al.* study and Experiment 1 was the age of the infants. In the earlier study infants ranged in age between about 8 and 15 weeks, while in Experiment 1 the infants were between 18 and 25 weeks of age. This raises the possibility that the infants in Experiment 1 may have responded differently than Coe *et al.*'s infants, not because of a difference in rearing conditions, but because of an age-related change in the distressfulness of separation.

In Experiment 2 we examined these different alternatives. To simulate an aunting situation, infants were provided with either a freshly laundered surrogate or a surrogate containing the scent of another infant monkey during separation. Olfactory cues play a key role in discrimination of the maternal figure by infant squirrel monkeys [19] and we have previously demonstrated that on the basis of such cues infants can discriminate their own surrogate from both a clean surrogate [18] and one belonging to a different infant [16]. If substitution of either of these surrogates following separation from the rearing surrogate was found to result in a greater cortisol response than separation followed by reunion with the rearing surrogate, it would suggest that the presence of an aunt or maternal substitute can, in fact, stimulate adrenocortical activity, even under conditions where clinging, a presumed reducer of distress [23], occurs. In addition, younger infants were used in Experiment 2 to insure that the findings of Experiment 1 did not reflect an ontogenetic change in the response to separation. Further, for purposes of comparison with the hormonal data, we also observed the behavior of the infants in response to removing the surrogate from the home cage or replacing it with a substitute. Our previous research with the squirrel monkey [4, 22, 25] has not examined the relationship of behavioral and cortisol changes to separation and such information would be helpful in the interpretation of results.

METHOD

Animals

The animals in this experiment were three male and seven female squirrel monkeys of the Bolivian variety. Rearing procedures were identical to those used in Experiment 1 and the infants were between 9 and 11 weeks of age at the time testing began.

Experimental Conditions

Experimental conditions were again imposed at weekly intervals in a counterbalanced order. In four of the conditions (2 to 5) 0.5 ml of blood was obtained 30 min after the manipulation was imposed, as in Experiment 1. The conditions were:

(1) Basal. The infant was simply removed from its cage, quickly anesthetized and its blood sampled as in the *Basal* condition of Experiment 1.

(2) Separated. This condition was identical to the *Separated-Familiar Environment-Nonhandled* condition of Experiment 1. The infant's surrogate was removed from the home cage with as little disturbance as possible to the infant.

(3) Clean Surrogate Substitute. In this condition the in-

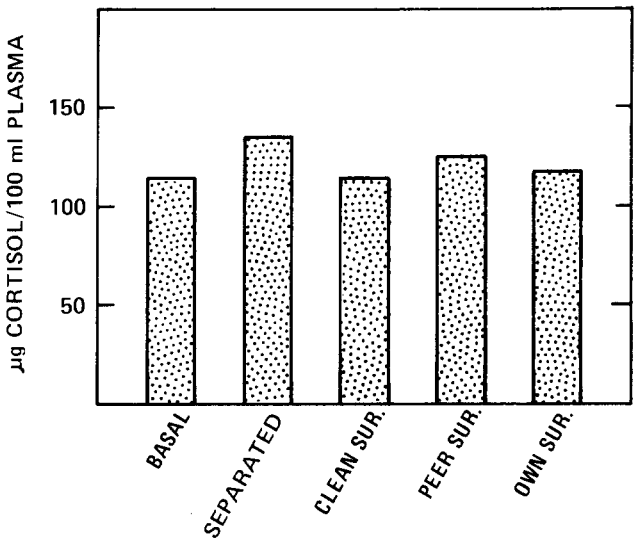


FIG. 2. Plasma cortisol levels of infant squirrel monkeys in each of the five experimental conditions of Experiment 2: *Basal; Separated; Clean Surrogate Substitute* (CLEAN SUR.); *Peer Surrogate Substitute* (PEER SUR.); *Own Surrogate* (OWN SUR.).

fant's surrogate was removed from the home cage, disturbing the infant as little as possible, and immediately replaced with a substitute surrogate that was identical in all respects, except that the cover had been freshly laundered. Infants had limited previous exposure to clean surrogates since their own surrogate had been laundered at weekly intervals.

(4) Peer Surrogate Substitute. In this condition the infant's surrogate was removed, disturbing the infant as little as possible, and immediately replaced with a surrogate that belonged to another infant. The substitute surrogate was identical to the infant's own surrogate except that the cover had not been washed for 4 to 6 days and thus contained accumulated odors deposited by the other infant. The infants had not been previously exposed to a surrogate containing the odors of another infant.

(5) Own Surrogate. In this condition the infant's surrogate was removed from the home cage and then immediately returned according to the same procedures used in the conditions in which substitute surrogates were provided.

Behavioral Observations

In Conditions 2 to 5, the infants' behavior was recorded during the first, second, and last 5-min segments of the 30-min period between the experimental manipulation and sampling of blood. Since no systematic changes in behavior occurred over the three 5-min observation periods, the data for these periods were pooled for statistical analyses. Behavioral measurements included: (1) Shrill Vocalization—the number of high-pitched shrill vocalizations; (2) Locomotor Activity—the number of crossings over the midline of the cage; (3) Environmental Manipulation—duration of oral and tactual contact with parts of the cage environment other than the surrogate; (4) Sniff Surrogate—frequency of sniffing the surrogate cover; and (5) On Surrogate—duration of time spent on the surrogate. The last two measures were recorded only for Conditions 3 to 5.

RESULTS

A one-way analysis of variance for repeated measures revealed no significant difference among conditions, $F<1$ (Fig. 2).

Behavior

Analyses of variance for repeated measures revealed significant effects for each behavior: Shrill Vocalization, $F(3,27)=4.62$, $p=0.01$; Locomotor Activity, $F(3,27)=15.35$, $p<0.001$; Environmental Manipulation, $F(3,27)=7.20$, $p<0.01$; Sniff Surrogate, $F(2,18)=8.21$, $p<0.01$; On Surrogate, $F(2,18)=15.23$, $p<0.001$. For each measure pairs of means were then compared using Newman-Keuls tests. The *Separated* condition differed from all other conditions on the measures of Shrill Vocalizations and Locomotor Activity. Infants emitted more high-pitched vocalizations ($ps<0.05$) and crossed the midline of the cage more frequently ($ps<0.01$) during the *Separated* condition (Fig. 3). Further

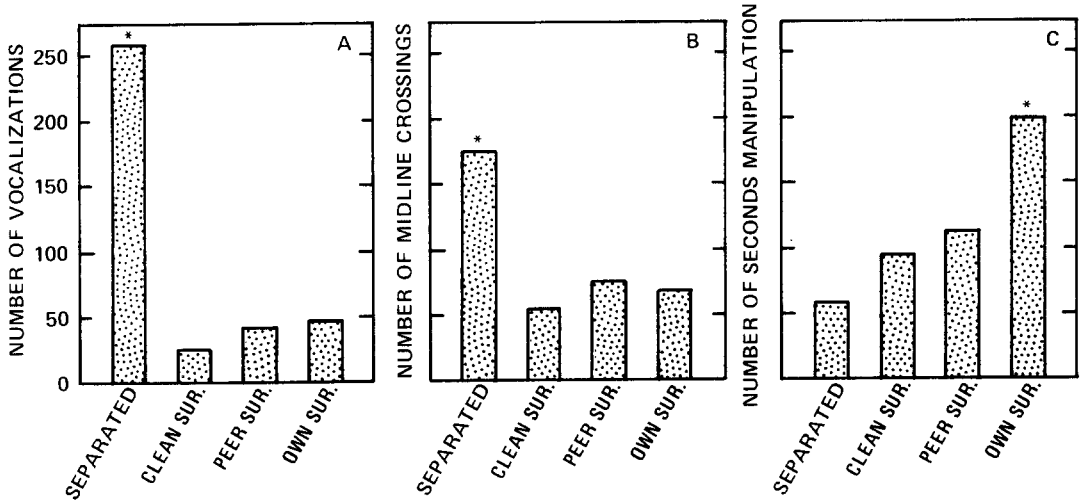


FIG. 3. Number of vocalizations, midline crossings and seconds of environmental manipulation by infants in Experimental Conditions 2 to 5 in Experiment 2. Abbreviations are the same as in Fig. 2. *Differs from all other means, $ps<0.05$ (A, C); $ps<0.01$ (B).

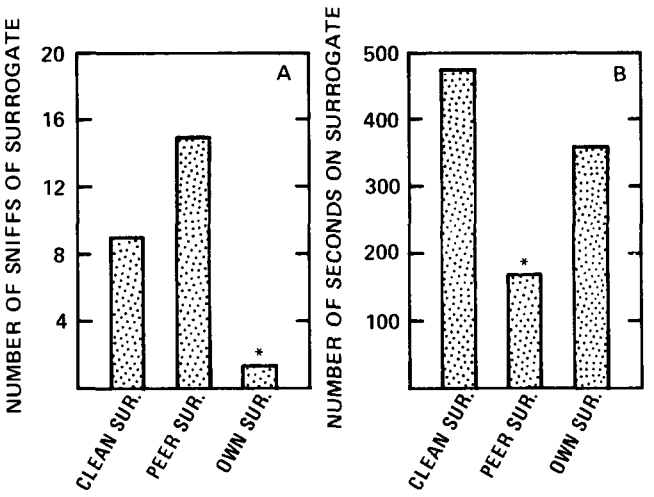


FIG. 4. Number of sniffs of surrogate and seconds spend on surrogate by infants in Experimental Conditions 3 to 5 in Experiment 2. Abbreviations are the same as in Fig. 2. *Differs from all other means, $ps<0.05$ (A); $ps<0.01$ (B).

analysis of Environmental Manipulation showed that when their own surrogate was returned, infants manipulated environmental objects more than they did following surrogate separation or presentation of either of the substitute surrogates ($ps<0.05$; Fig. 3). Infants also sniffed their own surrogate less than they did the substitute surrogates ($ps<0.05$; Fig. 4). Finally, the infants spent significantly less time on the surrogate of a peer than they did on their own surrogate or a clean surrogate ($ps<0.02$; Fig. 4).

GENERAL DISCUSSION

In Experiment 2, we tested infants of an age comparable to that used in our earlier work [4, 22, 25] and found no increase in plasma cortisol concentrations following removal of the surrogate from the infant's cage. This was true regardless of whether or not a differently scented substitute surrogate was provided during the separation period. Evidence that the substitute surrogates were in fact discriminated from the rearing surrogate, as well as from each other, is provided by the differential sniffing and time spent on the surrogates as well as by the relative amounts of environmental manipulation observed in the three surrogate conditions. Behavioral responses to separation were also found. Separation with no surrogate substitution resulted in increases in shrill vocalizations and activity. Moreover, infants separated from the rearing surrogate showed a suppression of environmental manipulation even if they were provided with a substitute surrogate during the separation period. Thus in terms of behavior alone, one might conclude that the surrogate-reared infants were reacting to brief separation by showing signs of protest or agitation just as mother-reared infants have been observed to do [12,13]. However, when the cortisol measure is also considered, discrepancies in the responses of surrogate-reared and mother-reared infants become apparent. Surrogate-reared infants show an elevation in plasma cortisol levels to separation only if the procedure involves removing the infant to an unfamiliar environment as seen in Experiment 1. This results differs from that obtained with mother-reared squirrel monkey infants which show an elevation in circulating cortisol levels even if the infant remains in the familiar home cage during the separation period [4].

An increase in plasma levels of adrenocorticoids is commonly observed in a wide range of distressful settings, and appears to be best considered as a physiological measure of the arousal of the responding animal [8]. Thus, the data of the present study together with those of Coe *et al.* [4] indicate that separation from the rearing surrogate induces less arousal than separation from the mother in infant squirrel monkeys. These findings are in agreement with previous results, which indicate that, in Old World macaques, separation is also more stressful for mother-reared than it is for surrogate-reared infants [26,29].

While it appears clear that attachment to the surrogate differs from attachment to the normal monkey mother, the precise manner in which it differs cannot be discerned from the present data. There appear to be at least three viable alternatives. The first possibility is that the attachment of the surrogate-reared infant to its surrogate is not as strong as the attachment of the mother-reared infant to its mother. According to this interpretation, removing the surrogate produced behavioral signs of agitation, but was not sufficiently arousing to induce an increase in pituitary-adrenal activity. The increased concentrations of plasma cortisol observed when the infant was separated and then placed into an unfamiliar environment were due solely to the unfamiliarity of that environment. Such unfamiliarity is known to be a potent elicitor of adrenal corticosteroid release [5,9]. In contrast to the surrogate-reared infant, the mother-reared squirrel monkey not only shows behavioral responses to separation [12,13] but also responds with a dramatic increase in plasma cortisol levels, even when the infant is left in the home cage with a maternal substitute during the separation period [4]. Thus, this line of reasoning would lead to the conclusion that the surrogate-reared infant shows less emotional reaction to separation, and therefore less attachment to the rearing figure, than does its mother-reared counterpart.

A second possibility is that the surrogate-reared infant becomes attached to familiar aspects of the environment in addition to its surrogate. According to this position, the surrogate does not offer as optimal conditions for attachment, or is as distinctly discriminable from surrounding environmental objects as is the biological mother (the point at which the inanimate surrogate ends and the rest of the home cage begins may be easier for the experimenter to distinguish than it is for the infant monkey). The surrogate-reared infant therefore shows attachment to various familiar features of the environment in addition to the surrogate, such as, perhaps, the home cage and possibly even surrounding infants and the ambient stimulation in the room. Removal of the surrogate thus resulted in only partial removal of objects to which the infant was attached. Under these conditions, the infant responded only with mild agitation as reflected in its behavior. This 'partial separation' was not sufficient to induce a state of heightened arousal or the concomitant increased secretion of cortisol from the adrenals. However, when the infant was removed to another room, it was separated from all attachment objects and in this case showed the typical cortisol response to separation.

A third interpretation is that the surrogate-reared infant shows no attachement. According to this view, there was no cortical response to separation. The increase in cortisol concentrations observed when infants were removed from the rearing room during the separation period is interpreted as evoked by the unfamiliarity of the environment into which

the infant was placed. Further, this position would argue that there was no behavioral response to separation from an attachment figure. Of the three behavioral measures which showed changes during separation, two of these, Shrill Vocalizations and Locomotor Activity, showed no change when the infant was provided with a discriminable maternal substitute during the separation period. Thus, it could be argued that vocalizations and locomotion increased in response to the absence of a soft surface upon which to cling, rather than to the absence of a specific attachment object. The third behavioral change observed during separation, suppressed Environmental Manipulation, did occur when substitute surrogates were provided, and this would appear to be an indication of separation distress from a specific attachment object. Yet, it should be pointed out that considerably more sniffing of the surrogate was observed under the two substitute surrogate conditions that when the infant was returned to its own surrogate. Therefore, the suppression of environmental manipulation seen in the two substitute conditions may have been the result of a concomitant increase in an incompatible behavior, surrogate sniffing, rather than a response to separation from an attachment figure. If this view is taken, there is no hormonal or behavioral response to separation by the surrogate-reared infants. Since distress upon involuntary separation is a defining characteristic of attachment [1], this line of reasoning would lead to the conclusion that surrogate-reared infants show no attachment.

Regardless of which of the above viewpoints is taken, the question remains as to why the surrogate does not afford the same favorable conditions for infant attachment as does the monkey mother. It has been suggested that attachment in monkeys develops as a result of the infant clinging to the attachment object [6,24]. However, the surrogate provides ample opportunity for clinging and infant squirrel monkeys do spend large amounts of time clinging to their surrogates [14]. It seems, therefore, that there must be some other factor upon which the surrogate and natural mother differ, which is critical for the development of infant attachment. One difference which may be of importance concerns the contingency between infant behaviors and the responses by the attachment figure. While the surrogate provides the infant with warmth, nourishment and a soft surface upon which to cling, it is essentially unresponsive to the infant. In contrast, the natural mother continuously responds to stimuli from the infant and adjusts her behavior to its changing demands. We suggest that the element of response contingency may be a necessary condition for normal attachment to occur.

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